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# Neural Morphogen Operators: Learning Continuous Reaction–Diffusion Dynamics from Spatial Transcriptomics

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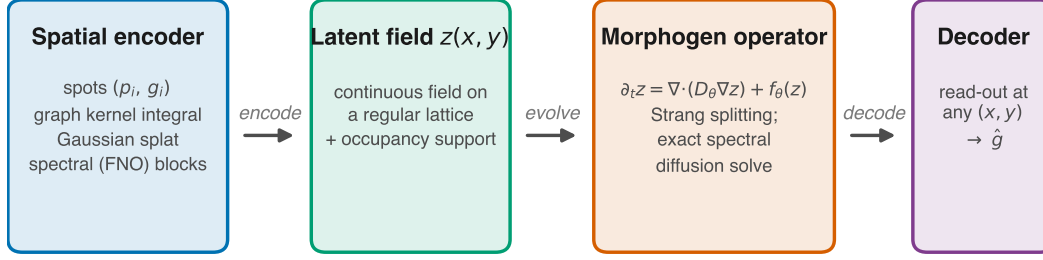
## Abstract

Spatial transcriptomics is almost universally modelled as a graph over measured spots, an inductive bias that ties the learned function to one sampling lattice and admits no notion of expression between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. This work introduces **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE  $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$ , and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting removes the CFL constraint and renders  $\mathbf{D}_\theta$  as a physical length scale (374  $\mu\text{m}$  median). On 9 sections from 6 public datasets spanning four technologies, NMO improves masked-region reconstruction over graph and Gaussian-process baselines ( $r = 0.247 \pm 0.005$  vs.  $0.234 \pm 0.001$ ). Two controls locate the effect: an exactly parameter-matched ablation shows the 0.43% of weights forming the operator account for  $-0.023$  Pearson  $r$ , while a non-spatial baseline shows graph message passing adds at most 0.005 over distance-weighted interpolation against 0.017 for the operator. Three further tests are negative or null: the fitted operator is dissipative with no Turing instability at any finite wavenumber, its fields are the most over-smoothed of any model tested, a pre-specified split-half perturbation test does not reach significance, and it transfers *worse* than the baseline it beats in-domain ( $2.8\text{--}2.0\times$ ). The continuous-operator view buys in-domain accuracy by specialising to a single tissue; its portability and mechanistic readings remain unsupported.

## 1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field: morphogens are produced, diffuse, decay and interact, and cells read the resulting landscape [Turing, 1952, Wolpert, 1969,



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 1: **Neural Morphogen Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure must pass through the field and therefore through the operator.

33 Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from  
 34 digit patterning [Sheth et al., 2012] to Nodal/Lefty signalling [Müller et al., 2012], with measured  
 35 gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue  
 36 organisation is generated by a continuous process, a model of that process—not of the lattice it was  
 37 sampled on—is the natural object to fit.

38 The proposition is tested directly. **Neural Morphogen Operators** (NMO) encode irregularly sam-  
 39 pled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion  
 40 PDE, and decode at arbitrary coordinates (Fig. 1). The question addressed is where this helps and  
 41 where it does not; the stronger question of whether the fitted operator recovers biochemistry is delib-  
 42 erately *not* posed, since a single snapshot cannot support it—a claim made precise in Theorem 14.

43 **Contributions.** (i) A PDE-constrained neural operator for spatial omics coupling a graph kernel-  
 44 integral encoder, a spectral reaction–diffusion operator and a coordinate-free decoder (Section 3).  
 45 (ii) An unconditionally stable, interpretable integrator: diffusion applied exactly in the Fourier do-  
 46 main under Strang splitting, removing the CFL constraint and rendering  $\mathbf{D}_\theta$  as a diffusion length in  
 47 microns. (iii) A correct formulation of the physics loss—because the integrator is exact the usual  
 48 PDE residual is vacuous, and the content-bearing constraint is a *steady-state* residual. (iv) Two con-  
 49 trols that locate the effect: an exactly parameter-matched  $T=0$  ablation, and a non-spatial baseline  
 50 showing how little graph message passing contributes. (v) Four negative results reported as results—  
 51 no Turing instability, worst-in-class spatial autocorrelation, a failed split-half perturbation test, and  
 52 worse transfer than the baseline surpassed in-domain.

## 53 2 Related Work

54 **Neural operators** learn maps between function spaces, giving approximate independence from the  
 55 discretisation [Kovachki et al., 2023]: Fourier neural operators parameterise the kernel spectrally [Li  
 56 et al., 2021], graph neural operators integrate a kernel over neighbourhoods on irregular meshes [Li  
 57 et al., 2020], DeepONet factorises into branch and trunk networks [Lu et al., 2021], and message-  
 58 passing solvers learn the numerical stencil itself [Brandstetter et al., 2022]. The graph form is used  
 59 for irregular sampling and the spectral form for the dynamics, the latter rendering the diffusion solve  
 60 exact; the integrator follows classical exponential time-differencing and operator splitting [Cox and  
 61 Matthews, 2002, Strang, 1968] rather than soft PDE-residual penalties [Raissi et al., 2019] or neural  
 62 ODEs [Chen et al., 2018].

63 **Machine learning for spatial transcriptomics** is dominated by graph models built for domain  
 64 identification and denoising rather than prediction at unmeasured coordinates—SpaGCN [Hu et al.,  
 65 2021], STAGATE [Dong and Zhang, 2022], SpatialDE/SPARK [Svensson et al., 2018, Sun et al.,

2020], DestVI [Lopez et al., 2022]—which is why they are adapted here (Section 4); Spateo [Qiu et al., 2024] is closest in spirit but targets 3-D reconstruction rather than a transferable operator. **Pattern formation** supplies the vocabulary [Turing, 1952, Kondo and Miura, 2010, Green and Sharpe, 2015]; RNA velocity [Bergen et al., 2020] and perturbation models [Lotfollahi et al., 2019, Roohani et al., 2024] model dynamics or interventions but carry no explicit spatial operator—the gap addressed here. Reaction–diffusion formulations have also been used to analyse over-smoothing in graph networks [Choi et al., 2023, Chamberlain et al., 2021], a connection the present results bear on directly.

### 3 Method

A section provides  $N$  measurements  $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$  with positions  $\mathbf{p}_i \in \mathbb{R}^2$  (isotropically normalised to  $[-1, 1]^2$ ) and expression  $\mathbf{g}_i \in \mathbb{R}^G$ . A latent field  $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$  is posited and the model factored as encode–evolve–decode.

**Spatial encoder.** Irregular sampling is handled by kernel-integral layers on a  $k$ -NN graph [Li et al., 2020],  $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$ , where  $\kappa_\theta$  maps a *relative* displacement to a per-channel gain; depending on geometry rather than node identity is what lets an encoder fitted to one lattice be evaluated on another. Features are splatted onto a regular  $H \times W$  lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map that distinguishes “tissue with no signal” from “no measurement here” — the distinction the masked-region task hinges on. Spectral blocks [Li et al., 2021] then refine the field.

**The morphogen operator.** The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

with a per-channel symmetric positive-definite tensor  $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$  ( $L_c$  lower triangular, so definiteness is structural rather than penalised) and a pointwise reaction network  $f_\theta$  implemented as  $1 \times 1$  convolutions with a tanh-bounded output.

Equation (1) is stiff—the diffusion operator has eigenvalues scaling as  $-|\mathbf{k}|^2$ , so an explicit scheme needs  $\Delta t = O(h^2/\|\mathbf{D}\|)$ —but for constant-coefficient anisotropic diffusion it is diagonal in the Fourier basis, so evolution over  $\tau$  has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

Composing (2) with a midpoint reaction update under Strang splitting [Strang, 1968] yields the one-step map  $S_{\Delta t} = e^{\frac{\Delta t}{2} \mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2} \mathcal{A}}$ , which is second-order accurate (Proposition 11). Two structural guarantees follow; both are proved in Appendix E and verified numerically in Table 2.

**Proposition 1** (Unconditional stability, no CFL condition). *For every  $\tau \geq 0$  and every  $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ ,  $\|e^{\tau \mathcal{A}_\mathbf{D}} u\|_{L^2} \leq \|u\|_{L^2}$ , with equality only for spatially constant  $u$ . Moreover the spatial mean is exactly conserved,  $\int_\Omega e^{\tau \mathcal{A}_\mathbf{D}} u = \int_\Omega u$ .*

Definiteness is structural rather than penalised ( $\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$ , Proposition 6), so Proposition 1 holds along the entire optimisation trajectory: no step size couples to learned parameters and no projection is required. Combined with  $\|f_\theta\|_\infty \leq \|g\|_\infty$  it yields a uniform absorbing set for the zero-mean component of radius  $\Delta t \|g\|_\infty |\Omega|^{1/2} / (1 - \rho)$ ,  $\rho = e^{-\Delta t \lambda_{\min} \pi^2}$  (Proposition 12); rollouts remain bounded for  $10^3$  steps empirically. Finally  $\mathbf{D}_c$  is interpretable: its eigenvalues are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

**Decoder.** Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input:* given coordinates it could memorise a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

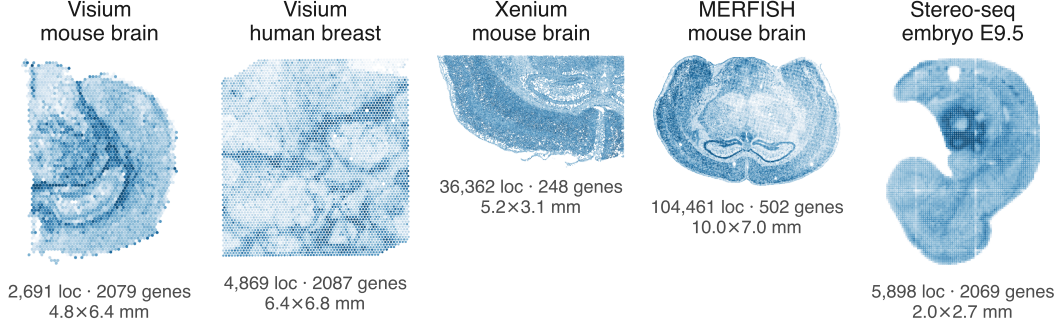


Figure 2: **Processed spatial sections**, coloured by total counts per location. The five sections differ in assay, sampling density (2.7k Visium spots to 104k MERFISH cells), gene panel (248–2087) and physical extent (2–10 mm), which is precisely what the transfer experiments vary. Full inventory in Appendix M.

**Objectives, and what a physics loss can mean here.** The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and three physics terms. The conventional physics-informed penalty—the PDE residual along the model’s own trajectory—is however vacuous for an exactly integrated operator.

**Proposition 2** (The trajectory residual is an integrator diagnostic). *Let  $R_{\Delta t}(\mathbf{z}; \theta) := \Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_{\theta}(\mathbf{z}))$ . Then  $\|R_{\Delta t}\| = \frac{\Delta t}{2}\|[\mathcal{A}_{\mathbf{D}}, \mathcal{F}_{\theta}]\mathbf{z}\| + O(\Delta t^2)$ , uniformly over bounded parameter sets, and  $R_{\Delta t}$  is independent of the observations.*

Minimising  $\|R_{\Delta t}\|^2$  therefore cannot fit the model to data—the data do not appear—and merely biases  $\theta$  towards operators whose diffusive and reactive parts commute, with a gradient signal vanishing as  $O(\Delta t)$  (Appendix F; observed order 1.00 in Table 2). The constraint that does carry content is that the *measured configuration be a near-stationary state* of the learned operator:

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_{\theta} \nabla \mathbf{z}_T) + f_{\theta}(\mathbf{z}_T)\|^2, \quad (3)$$

the correct formalisation of the quasi-steady-state assumption for adult tissue, constraining  $\mathbf{D}_{\theta}$  and  $f_{\theta}$  jointly. Mass, Jacobian-norm and splitting-consistency terms are also included; the ablations of Section 6 show all three to be inert at the weights adopted here.

**Interpretability: dispersion analysis.** Linearising (1) about a homogeneous  $\mathbf{z}^*$  gives  $\partial_t \delta \mathbf{z}_{\mathbf{k}} = (\mathbf{J} - \mathbf{k}^T \mathbf{D} \mathbf{k}) \delta \mathbf{z}_{\mathbf{k}}$  with  $\mathbf{J} = \partial f_{\theta} / \partial \mathbf{z}|_{\mathbf{z}^*}$ , so  $\max \text{Respec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$  against  $|\mathbf{k}|$  is the dispersion relation of the *learned* operator (maximised over orientations of  $\mathbf{k}$ , since  $\mathbf{D}$  is anisotropic). Positive growth at finite  $|\mathbf{k}|$  with decay at  $\mathbf{k} = 0$  is the classical Turing signature: a falsifiable test rather than a post-hoc visualisation.

## 4 Experimental setup

**Data.** 6 public datasets span four spatial technologies plus one non-spatial perturbation screen (9 sections, 469,038 locations; Fig. 2). Raw reads are never used: vendor count matrices are consumed directly and a SHA256 manifest is recorded per artifact. One QC policy is applied throughout, then library-size normalisation, log transform, and HVG selection that *force-includes* every detected morphogen-pathway gene, so the interpretability analysis is not evaluated on a set selection already biased. Coordinates are normalised **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned diffusion coefficient meaningless.

**Task and splits.** Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and every model predicts expression at those coordinates. Random spot-level splits leak,

Table 1: Masked spatial reconstruction on visium\_mouse\_brain. Contiguous tissue blocks are hidden from every model; metrics are computed on held-out locations only. Mean  $\pm$  s.d. over 3 seeds. All models share the identical training loop, masks and evaluation code.  $|\Delta I|$  is the mean absolute error in Moran’s  $I$  between predicted and measured expression maps: it penalises over-smoothing that correlation alone does not detect.

| model                     | params | Pearson $r \uparrow$                | Spearman $\rho \uparrow$            | RMSE $\downarrow$                   | SSIM $\uparrow$                     | $ \Delta I  \downarrow$             |
|---------------------------|--------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| SpaGCN-style              | 0.59M  | $0.234 \pm 0.001$                   | $0.184 \pm 0.001$                   | $0.254 \pm 0.001$                   | $0.393 \pm 0.001$                   | $0.738 \pm 0.002$                   |
| STAGATE-style             | 0.69M  | $0.234 \pm 0.002$                   | $0.185 \pm 0.002$                   | $0.254 \pm 0.001$                   | $0.394 \pm 0.002$                   | $0.737 \pm 0.003$                   |
| Graph Transformer         | 0.83M  | $0.230 \pm 0.000$                   | $0.182 \pm 0.001$                   | $0.255 \pm 0.001$                   | $0.390 \pm 0.000$                   | $0.723 \pm 0.004$                   |
| Autoencoder (non-spatial) | 0.58M  | $0.230 \pm 0.001$                   | $0.180 \pm 0.002$                   | $0.255 \pm 0.000$                   | $0.387 \pm 0.002$                   | $0.727 \pm 0.001$                   |
| GNN (GCN)                 | 0.61M  | $0.228 \pm 0.001$                   | $0.181 \pm 0.001$                   | $0.253 \pm 0.001$                   | $0.386 \pm 0.002$                   | $0.755 \pm 0.002$                   |
| Gaussian Process          | 0.00M  | $0.149 \pm 0.008$                   | $0.108 \pm 0.006$                   | $0.293 \pm 0.004$                   | $0.297 \pm 0.012$                   | <b><math>0.718 \pm 0.003</math></b> |
| NMO (ours)                | 2.20M  | <b><math>0.247 \pm 0.005</math></b> | <b><math>0.203 \pm 0.003</math></b> | <b><math>0.252 \pm 0.004</math></b> | <b><math>0.404 \pm 0.006</math></b> | $0.762 \pm 0.004$                   |

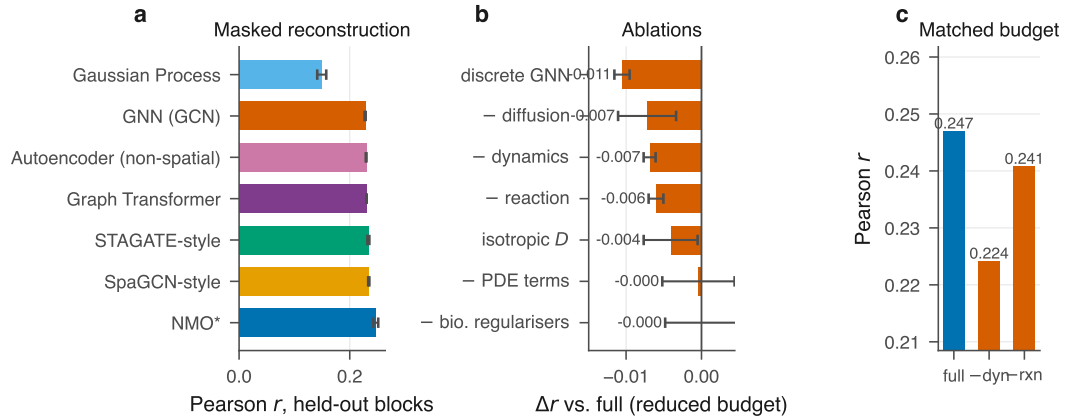


Figure 3: **Where the effect comes from.** (a) Masked-region reconstruction on the adult mouse-brain Visium section; NMO leads, but the graph baselines cluster within 0.007 of the *non-spatial* autoencoder. (b) Ablation deltas at the reduced budget: structural terms matter, regularisers do not. (c) The two decisive controls at the benchmark budget—disabling the operator ( $T=0$ ) costs more than NMO’s entire margin over the best baseline and drops it below every model in (a).

138 since a held-out spot is trivially predicted from its correlated neighbours, so whole contiguous blocks  
139 are held out [Roberts et al., 2017] and standardisation statistics are computed on the training split  
140 alone.

141 **Baselines.** A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style  
142 and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani  
143 et al., 2017], and an exact GP [Rasmussen and Williams, 2006]. The graph methods were not  
144 designed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out  
145 head—the most favourable choice available. These variants are marked *-style*; no claim is made that  
146 they reproduce the authors’ published numbers on the authors’ own tasks. All models share one  
147 training loop, masks, optimiser and evaluation code.

148 **Metrics.** Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al.,  
149 2004] on rasterised maps, and absolute error in Moran’s  $I$  [Moran, 1950], the last included because  
150 correlation and RMSE both reward regression towards a smooth mean. Mean  $\pm$  s.d. over 3 seeds.

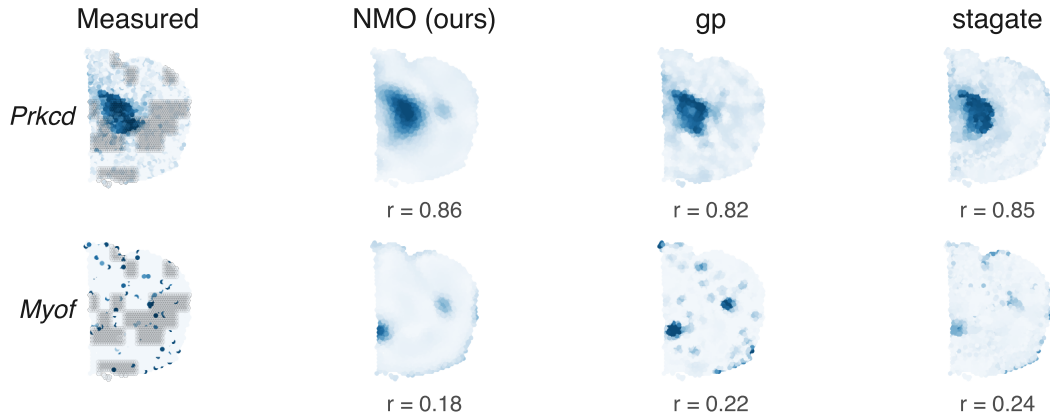


Figure 4: **Masked reconstruction across the performance range.** Measured expression (left; held-out spots outlined) against NMO and two baselines, for genes at the best, median and lower-quartile of per-gene Pearson  $r$ . Predicted fields are visibly smoother than the measurement—the bias that Moran’s  $I$  quantifies in Table 1.

## 5 Results

### 5.1 Masked spatial reconstruction

NMO reaches Pearson  $r$  of  $0.247 \pm 0.005$  on held-out blocks against  $0.234 \pm 0.001$  for the strongest baseline (spagcn), a gain of 0.013 (Table 1, Fig. 3a). The margin is modest and the graph baselines cluster tightly, so the ablations constitute the stronger evidence. The Gaussian process underperforms them—a property of this baseline rather than of GP regression, since a single RBF length scale is shared across all genes.

**Graph message passing barely beats plain interpolation.** The most informative row is the one included as a floor. The non-spatial autoencoder sees no coordinates at all and reaches 0.230 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates what *message passing* contributes. That contribution is close to nothing: across 4 graph models the margin spans  $-0.002$  to  $0.005$  Pearson  $r$ , and 1 is worse than using no spatial information at all, while NMO gains 0.017 — roughly  $3.6\times$  the best graph model’s margin. On this task the standard graph treatment extracts remarkably little beyond distance-weighted interpolation; the control merits routine reporting.

**One metric disagrees, and it is the one added as a guard.** NMO is best on Pearson, Spearman, RMSE and SSIM ( $0.404 \pm 0.006$  against 0.394), but *worst* on Moran’s  $I$  error:  $0.762 \pm 0.004$  against 0.718, a gap of  $+0.044$ . SSIM measures local contrast on a rasterised map while Moran’s  $I$  measures global autocorrelation on the point set, and NMO’s local structure is good while its fields are far too autocorrelated overall ( $I = 0.936 \pm 0.004$  against a measured 0.174; Fig. 4). The cause is the dissipative operator (Fig. 5c), which attenuates high spatial frequencies—helpful inside a masked block, harmful to texture.

### 5.2 Learned fields and dynamics

The fitted operator has a median diffusion length of  $374 \mu\text{m}$  (10–90th pct. 295–478), the same order as measured morphogen gradients [Kicheva et al., 2007, Yu et al., 2009] —a consistency check rather than a measurement of physical transport.

**The learned operator is not Turing-unstable.** The dispersion relation (Fig. 5c) decreases monotonically in  $|\mathbf{k}|$ : growth is weakly positive only for the uniform mode and strictly negative at every finite wavenumber, so no band of spatial modes is spontaneously amplified and no characteristic

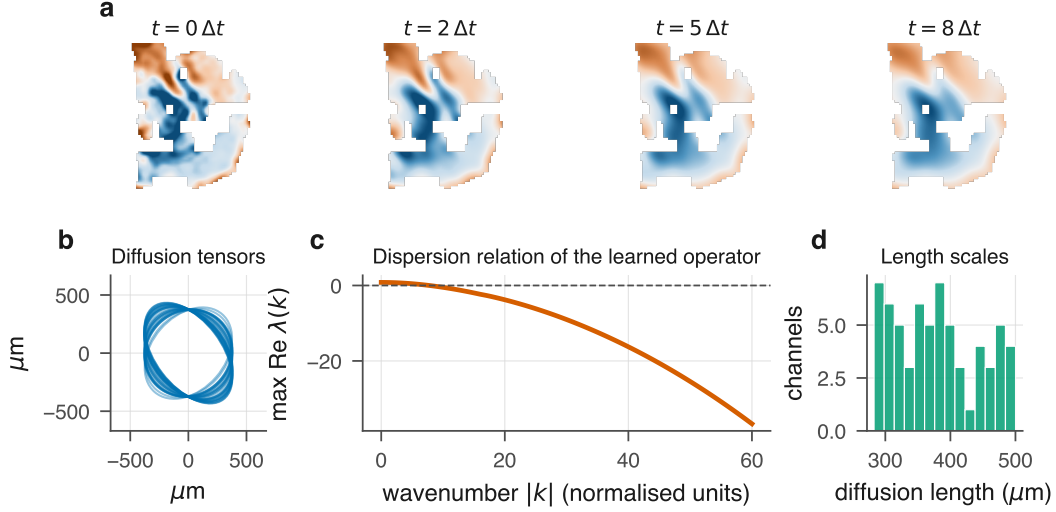


Figure 5: **What the fitted operator looks like.** (a) A latent channel relaxing under the learned dynamics. (b) Per-channel diffusion tensors as length-scale ellipses, showing the anisotropy is used. (c) Dispersion relation: growth is negative at every finite wavenumber, so the operator is dissipative and supports no Turing band. (d) Diffusion lengths across channels.

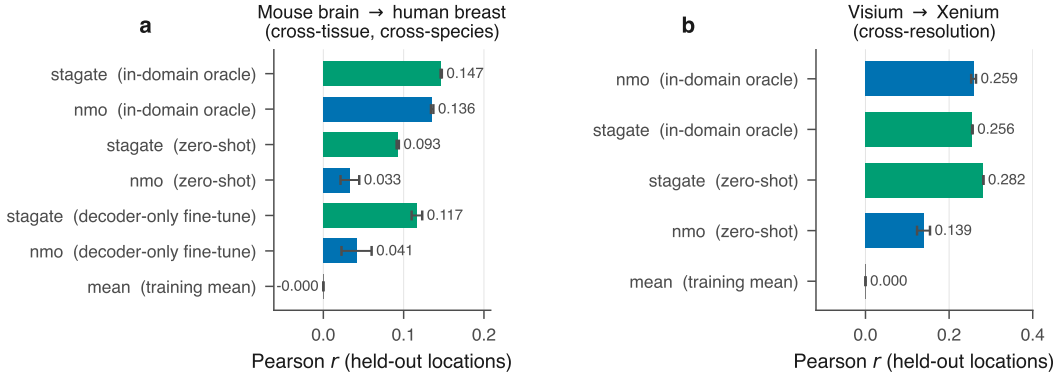


Figure 6: **Transfer is where the operator fails.** Zero-shot performance against the in-domain oracle and the training-mean floor, for (a) mouse brain  $\rightarrow$  human breast carcinoma and (b) Visium  $\rightarrow$  single-cell Xenium. The STAGATE-style baseline that NMO beats in-domain transfers substantially better in both settings.

180 wavelength exists. Fitted operators were Turing-unstable in 0% of runs. From a stationary snapshot  
 181 the model learns a *dissipative* operator whose fixed point is the observed tissue—exactly what (3)  
 182 asks for. The absence of a Turing band reflects the data available, not evidence against reaction-  
 183 diffusion accounts of the biology; detecting genuine pattern-forming instability would require ob-  
 184 serving tissue away from steady state.

### 185 5.3 Transfer across tissue, species and resolution

186 This is the clearest negative result in the paper. Trained on adult mouse brain and applied un-  
 187 changed to human breast carcinoma—a different species, organ and spatial architecture—NMO  
 188 attains  $0.033 \pm 0.012$  zero-shot against a training-mean floor of  $-0.000 \pm 0.000$  and an in-domain  
 189 oracle of  $0.136 \pm 0.001$ . It is above the floor, so something transfers. But stagate-style reaches  
 190  $0.093 \pm 0.001$  in the same setting,  $2.8\times$  better than the model that beat it in-domain (Fig. 6); re-  
 191 fitting only the decoder does not close the gap ( $0.041 \pm 0.019$ ). Cross-resolution repeats the pattern: an  
 192 operator fitted to  $55 \mu m$  Visium spots and evaluated on single-cell Xenium sampling of the *same or-*

193 *gan* reaches  $0.139 \pm 0.015$  against  $0.282 \pm 0.000$ , again  $2.0 \times$  worse (Appendix P). All Xenium panel  
 194 genes are present in the raw Visium matrix, so the source is rebuilt with the panel force-included  
 195 (248 shared genes); the gap is not an artefact of thin gene overlap.

196 The opposite was anticipated, since an operator defined on the field rather than on a lattice should  
 197 tolerate changes of geometry. The most plausible account—advanced as a hypothesis rather than a  
 198 demonstration—is that the advantage and the failure share a cause. The in-domain gain is obtained  
 199 by fitting an operator whose fixed point *is* the source tissue; relaxation towards that attractor is  
 200 actively wrong when the target has a different architecture, whereas local message passing encodes  
 201 less about the source and degrades more gracefully. Appendix I makes the underlying distinction  
 202 explicit: Proposition 20 establishes discretisation invariance of the *encoder*, which does not entail  
 203 portability of the *fitted* operator, and Theorem 14 indicates why.

## 204 5.4 Developmental forecasting

205 On the Stereo-seq series—the only dataset here with a genuine temporal axis—integrating the en-  
 206 coded E9.5 field forward improves the field-level match to E10.5 from 0.082 at  $T=0$  to 0.095 at  
 207  $T=16$ , against 0.064 for predicting no change. The learned operator therefore carries genuine tem-  
 208 poral information, with an interior optimum in  $T$ . Consecutive MOSTA stages are distinct embryos,  
 209 so the comparison is at the level of the rasterised field after coarse registration (Appendix R).

## 210 5.5 Counterfactual perturbation

211 The pathway interpretation is tested by a falsifiable split-half design: the latent direction maximally  
 212 raising one random half of a morphogen pathway is injected, the operator integrated, and the re-  
 213 sponse of the *held-out* half assessed against a permutation null over matched gene sets.

214 **It fails.** Across 4 pathways and 2 seeds, 0 reach significance (smallest  $p = 0.47$ ) and the mean  
 215 held-out rank is 59% against a 50% chance level—held-out pathway members respond, if anything,  
 216 slightly *less* than random genes. This rules out the reading Fig. 7b would otherwise invite: the  
 217 correlations are consistent with the field encoding coarse anatomy that pathway scores also track.  
 218 The outcome is moreover anticipated by Corollary 15, since the perturbation evaluates  $f_\theta$  off the  
 219 observed manifold  $\mathcal{R}$  where the data impose no constraint. A scale is nonetheless established: the  
 220 response decays to half its peak within  $577 \mu\text{m}$ , matching the diffusion lengths of  $\mathbf{D}_\theta$ . It is the  
 221 *specificity*, not the range, that is rejected. A second pre-specified probe against Perturb-seq [Norman  
 222 et al., 2019] was underpowered ( $n = 7$ ) and is inconclusive (Appendix Q).

## 223 6 Ablations

224 The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is  
 225 never applied. The sweep (Fig. 3b, Appendix L) uses a reduced budget at which the full model has  
 226 *not* converged (0.233 against 0.247); since undertraining compresses every delta, the two decisive  
 227 variants were re-run at the benchmark budget (3 seeds; Fig. 3c), giving  $0.224 \pm 0.005$  ( $-0.023$ ) for  
 228  $T=0$  and  $0.241 \pm 0.003$  ( $-0.006$ ) without the reaction. The  $T=0$  value falls below *every* baseline  
 229 of Fig. 3a, including the non-spatial one.

230 The control is *exactly* parameter-matched: the operator is 9,536 of the model’s 2,198,336 parameters  
 231 (0.43%), and  $T=0$  leaves them allocated but unused, so architecture, parameter count and optimiser  
 232 budget are identical and the gap cannot be capacity. That matters because NMO is larger than the  
 233 graph baselines, so part of its Table 1 margin could be capacity; the  $T=0$  control separates those  
 234 explanations, and it is accordingly weighted above the margin.

235 Two ablations are honestly *null*: removing the biological regularisers changes held-out  $r$  by  $-0.000$   
 236 and removing the PDE constraint terms by  $-0.000$ , both within seed noise. At the weights adopted  
 237 here these terms are inert. The rows are retained rather than dropped, since a reader is entitled  
 238 to know that three of the five loss terms of Section 3 do no measurable predictive work, whatever



they contribute to the operator’s form. Constraining  $\mathbf{D}$  to be isotropic costs -0.004, indicating the anisotropy is used rather than merely available.

## 7 Limitations

**One snapshot cannot identify a dynamical system.** This is not merely a practical caveat but a structural one. Writing  $\mathcal{R} := \mathbf{z}^*(\Omega) \subset \mathbb{R}^C$  for the range of the observed latent field, Theorem 14 states that for  $C \geq 3$  the stationarity constraint determines  $f_\theta$  only on  $\mathcal{R}$ , a set of Hausdorff dimension at most 2 and hence Lebesgue-null: the admissible set contains  $f_\theta + \varphi$  for every  $\varphi$  vanishing on  $\mathcal{R}$ , an infinite-dimensional family. Reported diffusion lengths therefore characterise the *fitted operator*, not morphogen transport, and any counterfactual that displaces  $\mathbf{z}$  off  $\mathcal{R}$ —as the perturbation of Section 5.5 does by construction—is fixed by the architecture and regularisers rather than by the data (Corollary 15). The null outcome reported there is thus predicted by the theory, not merely observed.

**Latent, not molecular; the name describes the operator class.** Latent/pathway correlations show alignment, but the split-half test of Section 5.5 rejects the stronger reading (Section 5.5). “Morphogen” names the *operator class*—anisotropic reaction–diffusion—not a validated claim about signalling. Two of three interpretability probes returned negative and a third was underpowered; what survives is a diffusion tensor with a physical length scale and a perturbation half-width consistent with it.

**Over-smoothing, worse than the baselines’.** Predicted maps are far more autocorrelated than measured ones ( $0.936 \pm 0.004$  vs  $0.174$ ): NMO is *worse* than every baseline on Moran’s  $I$  by  $+0.044$  while leading on SSIM by  $0.010$ . The dissipative operator that delivers the accuracy gain inflates the autocorrelation — one phenomenon, not two. Predicting expression at unmeasured coordinates is supported; delineating sharp boundaries, rare niches or tumour margins is not.

**The operator is tissue-specific.** Discretisation invariance motivated preferring an operator to a graph, yet under transfer the graph baseline wins by  $2.8\times$  and  $2.0\times$ ; note that Proposition 20 concerns the *encoder* alone. Until a shared operator is fitted across tissues, or  $\mathbf{D}_\theta$  conditioned on tissue covariates, this remains a within-section model.

**Adapted baselines and limited scope.** The *-style* variants implement each method’s architectural core with a common read-out head: fair on *this* task, but not a reproduction of published results. The in-domain margin is a few points of Pearson  $r$  on one section per condition over few seeds, without a power analysis, and scope is limited to single 2-D sections and CPU-scale compute.

## 8 Discussion and Conclusion

The question posed was whether the continuous description developmental biology uses for tissue can be fitted as an operator on spatial transcriptomics. The answer is a qualified affirmative with a sharp boundary. In-domain the formulation helps: the parameter-matched  $T=0$  control costs more than the entire margin over the best baseline, and on the one dataset with a real time axis integration beats persistence, while the exact spectral solve renders the operator stable and physically readable.

Outside that boundary the results are negative, and they are the more useful half of the paper: the operator does not transfer, it is dissipative rather than pattern-forming, its fields are the most over-smoothed of any model tested, and a falsifiable test of the pathway interpretation fails outright. The continuous-operator framing buys in-domain accuracy by specialising to one tissue; portability and mechanism remain unearned, and breaking the identifiability barrier of Theorem 14 requires dense temporal sampling or interventional spatial data.

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Table 2: Numerical verification of the propositions of Section 3 and Appendix E. Each analytical claim is checked against the implementation under adversarially perturbed parameters; the script that produces this table is released with the code.

| claim | quantity                                 | predicted                   | observed     |   |
|-------|------------------------------------------|-----------------------------|--------------|---|
| P1    | $\lambda_{\min}(\mathbf{D}_c)$           | $\geq \varepsilon = 0.0001$ | 1.000e-04    | • |
| P2    | $\sup_{\Delta t} \ E(\Delta t)u\ /\ u\ $ | $\leq 1$                    | 0.6575769247 | • |
| P3    | $\max_c  \Delta \int_{\Omega} u_c $      | 0 (exact)                   | 1.04e-17     | • |
| P4    | $\min_c (g_c - \ f_c\ _{\infty})$        | $\geq 0$                    | 0.00e+00     | • |
| P5    | observed order of $S_{\Delta t}$         | 2                           | 2.00         | • |
| P6    | $\sup_n \ z'_n\ $ vs. analytic bound     | $\leq 220.6$                | 68.5         | • |
| P7    | observed order of $\ R_{\Delta t}\ $     | 1                           | 1.00         | • |

## 381 A Latent fields

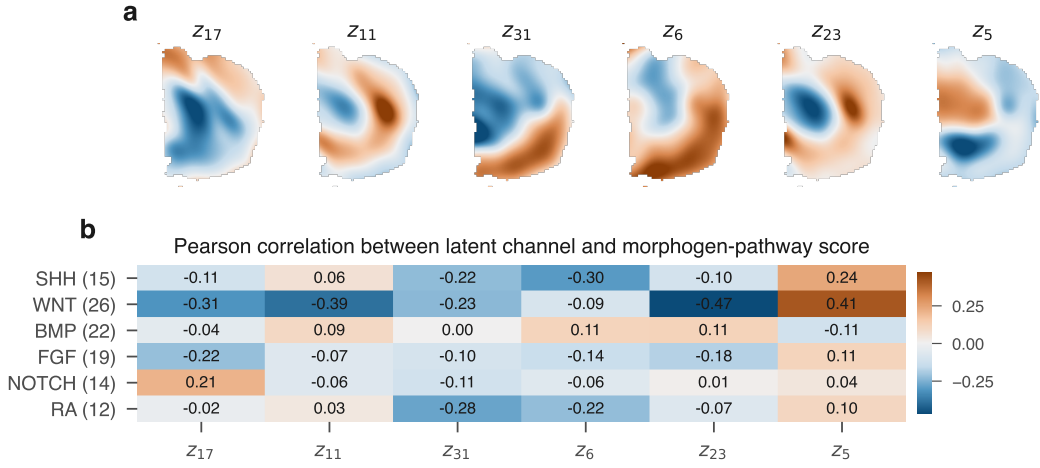


Figure 7: **Latent fields align with anatomy, not with pathways.** (a) Latent channels of the relaxed field recover laminar and nuclear structure. (b) Correlation between each channel and canonical morphogen-pathway scores. These correlations are real, but do not survive the causal test.

## 382 B Numerical verification of the analytical claims

383 Each proposition below is accompanied by an executable check; Table 2 reports the outcome under  
384 adversarially perturbed parameters (diffusion parameters drawn uniformly on  $[-60, 60]$ , reaction  
385 weights perturbed by  $\mathcal{N}(0, 2^2)$ ), so the guarantees are exercised well outside the regime reached  
386 during training.

## 387 C Notation and function-space setting

388 Throughout,  $\Omega = [-1, 1]^2$  is identified with the flat torus  $\mathbb{T}^2 = (\mathbb{R}/2\mathbb{Z})^2$ , so that the spectral  
389 operators below are exact rather than approximate. Let  $L^2(\Omega; \mathbb{R}^C)$  carry the inner product  $\langle u, v \rangle =$   
390  $\int_{\Omega} u(x)^\top v(x) dx$  and norm  $\|u\| = \langle u, u \rangle^{1/2}$ . The Fourier transform is

$$\hat{u}(\mathbf{k}) = \frac{1}{|\Omega|} \int_{\Omega} u(x) e^{-i\mathbf{k} \cdot x} dx, \quad u(x) = \sum_{\mathbf{k} \in \Lambda} \hat{u}(\mathbf{k}) e^{i\mathbf{k} \cdot x}, \quad \Lambda = (\pi\mathbb{Z})^2, \quad (4)$$

391 with Parseval identity  $\|u\|^2 = |\Omega| \sum_{\mathbf{k} \in \Lambda} |\hat{u}(\mathbf{k})|^2$ . The lowest non-zero wavenumber on  $\Omega$  is  
392  $|\mathbf{k}|_{\min} = \pi$ .

Let  $\mathcal{S}_{++}^2$  denote the cone of symmetric positive-definite  $2 \times 2$  matrices. A *diffusion tensor field* is a tuple  $\mathbf{D} = (\mathbf{D}_1, \dots, \mathbf{D}_C) \in (\mathcal{S}_{++}^2)^C$ , acting channelwise. The *reaction field* is a map  $f_\theta \in C^1(\mathbb{R}^C; \mathbb{R}^C)$  applied pointwise,  $(f_\theta u)(x) = f_\theta(u(x))$ .

**Definition 3** (Morphogen operator). For  $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$  and  $f_\theta \in C^1(\mathbb{R}^C; \mathbb{R}^C)$  the *morphogen operator* is the semilinear evolution equation

$$\partial_t \mathbf{z} = \mathcal{A}_{\mathbf{D}} \mathbf{z} + f_\theta(\mathbf{z}), \quad (\mathcal{A}_{\mathbf{D}} \mathbf{z})_c := \nabla \cdot (\mathbf{D}_c \nabla \mathbf{z}_c), \quad (5)$$

on  $\Omega$  with periodic boundary conditions and initial datum  $\mathbf{z}(\cdot, 0) = \mathbf{z}_0 \in L^2$ .

Equation (5) is the classical semilinear parabolic form: a strongly elliptic second-order generator plus a globally Lipschitz, bounded perturbation. Existence and uniqueness of a mild solution on  $[0, \infty)$  is standard, and is recalled as Proposition 10 for completeness.

## D Spectral representation of the anisotropic generator

**Lemma 4** (Fourier symbol). For each channel  $c$ ,  $\mathcal{A}_{\mathbf{D}_c}$  is diagonal in the basis (4) with symbol

$$\widehat{\mathcal{A}_{\mathbf{D}_c} u}(\mathbf{k}) = -q_c(\mathbf{k}) \hat{u}(\mathbf{k}), \quad q_c(\mathbf{k}) := \mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq \lambda_{\min}(\mathbf{D}_c) |\mathbf{k}|^2 \geq 0. \quad (6)$$

*Proof.* Writing  $\mathbf{D}_c = (d_{ij})$  and  $u = e^{i\mathbf{k} \cdot \mathbf{x}}$ ,  $\nabla u = i\mathbf{k} u$ , hence  $\mathbf{D}_c \nabla u = i\mathbf{D}_c \mathbf{k} u$  and  $\nabla \cdot (\mathbf{D}_c \nabla u) = i\mathbf{k} \cdot (i\mathbf{D}_c \mathbf{k}) u = -(\mathbf{k}^\top \mathbf{D}_c \mathbf{k}) u$ . The inequality is the Rayleigh bound for symmetric  $\mathbf{D}_c$ .  $\square$

Consequently the diffusion semigroup admits the closed form used in the implementation,

$$(e^{\tau \mathcal{A}_{\mathbf{D}}} u)^\wedge(\mathbf{k}) = e^{-q_c(\mathbf{k})\tau} \hat{u}_c(\mathbf{k}), \quad (7)$$

which is evaluated by a pair of FFTs at cost  $O(CHW \log HW)$  per half-step. No spatial discretisation of  $\nabla \cdot (\mathbf{D} \nabla \cdot)$  is ever formed, so the diffusive update carries no truncation error beyond the spectral representation of  $\mathbf{z}$  itself.

**Remark 5** (Why anisotropy is retained). Isotropic  $\mathbf{D}_c = d_c I$  gives  $q_c(\mathbf{k}) = d_c |\mathbf{k}|^2$ , a symbol invariant under rotations of  $\mathbf{k}$ . The general case has level sets that are ellipses with principal axes given by the eigenvectors of  $\mathbf{D}_c$ , which is what permits different decay lengths along and across an anatomical axis. The ablation of Section 6 measures whether this freedom is used.

## E Structural guarantees: proofs

**Proposition 6** (Structural positive-definiteness). Let  $L_c$  be lower triangular with  $\text{diag}(L_c) = \text{softplus}(\ell_c)$  and let  $\mathbf{D}_c := L_c L_c^\top + \varepsilon I$  with  $\varepsilon > 0$ . Then  $\mathbf{D}_c \in \mathcal{S}_{++}^2$  with  $\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$ , for every value of the unconstrained parameters.

*Proof.*  $L_c L_c^\top$  is symmetric positive semi-definite, so for any  $v \neq 0$ ,  $v^\top \mathbf{D}_c v = \|L_c^\top v\|^2 + \varepsilon \|v\|^2 \geq \varepsilon \|v\|^2$ . Symmetry is immediate. Hence the spectrum lies in  $[\varepsilon, \infty)$ .  $\square$

The significance is that definiteness is a property of the *parametrisation* rather than of the optimiser's trajectory: no penalty enforces it and no projection step is required, so it cannot be violated transiently during training.

**Proposition 7** (Unconditional  $L^2$  contraction). For every  $\tau \geq 0$  and every  $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ ,  $\|e^{\tau \mathcal{A}_{\mathbf{D}}} u\| \leq \|u\|$ , with equality iff  $u$  is spatially constant. In particular the diffusive half-step imposes no Courant–Friedrichs–Lewy restriction on  $\Delta t$ .

*Proof.* By Parseval and (7),  $\|e^{\tau \mathcal{A}_{\mathbf{D}}} u\|^2 = |\Omega| \sum_{\mathbf{k}, c} e^{-2q_c(\mathbf{k})\tau} |\hat{u}_c(\mathbf{k})|^2 \leq |\Omega| \sum_{\mathbf{k}, c} |\hat{u}_c(\mathbf{k})|^2 = \|u\|^2$ , since  $q_c(\mathbf{k}) \geq 0$  gives  $e^{-2q_c\tau} \in (0, 1]$ . Equality forces  $e^{-2q_c(\mathbf{k})\tau} = 1$  wherever  $\hat{u}_c(\mathbf{k}) \neq 0$ ; for  $\tau > 0$  this requires  $q_c(\mathbf{k}) = 0$ , and by Lemma 4 with  $\lambda_{\min} \geq \varepsilon > 0$  this holds only at  $\mathbf{k} = 0$ .  $\square$

Proposition 7 is the formal content of the stability claim. An explicit scheme applied to (5) is stable only for  $\Delta t \lesssim h^2/\lambda_{\max}(\mathbf{D})$ , a constraint that couples the step size to *learned* parameters and can therefore be violated mid-training as  $\mathbf{D}_\theta$  adapts. The exponential update removes the constraint entirely.

**Proposition 8** (Exact conservation of the mean).  $\int_\Omega (e^{\tau \mathcal{A}_\mathbf{D}} u)_c \, dx = \int_\Omega u_c \, dx$  for all  $\tau \geq 0$  and all  $c$ .

*Proof.* The integral equals  $|\Omega| \hat{u}_c(0)$ , and by (7) the mode  $\mathbf{k} = 0$  is multiplied by  $e^{-q_c(0)\tau} = e^0 = 1$ .  $\square$

Proposition 8 licenses the interpretation of the mass penalty in Section 3: because the diffusive half-step is exactly conservative, any drift in  $\int \mathbf{z}$  along a trajectory is attributable to  $f_\theta$  alone, and penalising  $\int_\Omega f_\theta(\mathbf{z})$  is therefore a statement about the reaction network rather than a numerical artefact.

**Proposition 9** (Uniform bound on the reaction). *Let  $f_\theta(u) = g \odot \tanh(h_\theta(u))$  with  $g \in \mathbb{R}_{>0}^C$ . Then  $\|f_\theta(u)\|_\infty \leq \|g\|_\infty$  for all  $u$ , and  $f_\theta$  is globally Lipschitz with constant  $L_f = \|g\|_\infty \text{Lip}(h_\theta)$ .*

*Proof.*  $|\tanh| \leq 1$  channelwise gives the first claim. For the second,  $\|\nabla_u f_\theta\| = \|\text{diag}(g) \text{diag}(1 - \tanh^2)\| \leq \|g\|_\infty \|\nabla_u h_\theta\|$  since  $0 < 1 - \tanh^2 \leq 1$ .  $\square$

**Proposition 10** (Well-posedness). *Under Proposition 9, (5) admits a unique mild solution  $\mathbf{z} \in C([0, \infty); L^2)$  for every  $\mathbf{z}_0 \in L^2$ .*

*Proof sketch.*  $\mathcal{A}_\mathbf{D}$  generates an analytic contraction semigroup by Proposition 7, and  $f_\theta$  is globally Lipschitz and bounded on  $L^2$  by Proposition 9. The variation-of-constants map  $\mathbf{z} \mapsto e^{t\mathcal{A}_\mathbf{D}} \mathbf{z}_0 + \int_0^t e^{(t-s)\mathcal{A}_\mathbf{D}} f_\theta(\mathbf{z}(s)) \, ds$  is a contraction on  $C([0, T]; L^2)$  for  $T$  small, and the bound of Proposition 9 precludes finite-time blow-up, so the local solution extends globally.  $\square$

**Proposition 11** (Second-order accuracy of the split step). *Let  $S_{\Delta t} := e^{\frac{\Delta t}{2}\mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2}\mathcal{A}}$ , where  $R_{\Delta t}$  is the explicit midpoint update for  $\dot{\mathbf{z}} = f_\theta(\mathbf{z})$ . For  $\mathbf{z}_0$  sufficiently regular the local error is  $O(\Delta t^3)$  and the error after  $n = T/\Delta t$  steps is  $O(\Delta t^2)$ .*

*Proof sketch.* Strang splitting of  $e^{\Delta t(\mathcal{A} + \mathcal{F})}$  has local error  $\frac{\Delta t^3}{24}([\mathcal{A}, [\mathcal{A}, \mathcal{F}]] - 2[\mathcal{F}, [\mathcal{F}, \mathcal{A}]]) + O(\Delta t^5)$  by the Baker–Campbell–Hausdorff expansion, and the midpoint rule contributes a further  $O(\Delta t^3)$  local error. Summation over  $n$  steps with the stability of Proposition 7 gives the global rate.  $\square$

The predicted rate is confirmed numerically in Table 2, which reports an observed order of 2.00 over a four-fold refinement.

**Proposition 12** (Uniform bound on the fluctuation component). *Write  $\mathbf{z} = \bar{\mathbf{z}} + \mathbf{z}'$  where  $\bar{\mathbf{z}}$  is the spatial mean and  $\mathbf{z}'$  has zero mean. Let  $\rho := e^{-\Delta t \lambda_{\min}(\mathbf{D})\pi^2} < 1$  and  $L := \Delta t \|g\|_\infty |\Omega|^{1/2}$ . Then the iterates  $\mathbf{z}_{n+1} = S_{\Delta t} \mathbf{z}_n$  satisfy*

$$\|\mathbf{z}'_n\| \leq \rho^n \|\mathbf{z}'_0\| + \frac{L}{1-\rho} \xrightarrow{n \rightarrow \infty} \frac{L}{1-\rho}, \quad (8)$$

and  $\|\bar{\mathbf{z}}_n\| \leq \|\bar{\mathbf{z}}_0\| + nL$ .

*Proof.* On the zero-mean subspace every active wavenumber satisfies  $|\mathbf{k}| \geq \pi$ , so by Lemma 4 the two half-steps contract  $\mathbf{z}'$  by at least  $e^{-\Delta t \lambda_{\min} \pi^2} = \rho$  in total. The reaction step adds at most  $\Delta t \|f_\theta\| \leq L$  in  $L^2$  by Proposition 9. Hence  $\|\mathbf{z}'_{n+1}\| \leq \rho \|\mathbf{z}'_n\| + L$ , and (8) follows by summing the geometric series. The mean is untouched by diffusion (Proposition 8) and receives at most  $L$  per step.  $\square$

Proposition 12 is stronger than mere absence of blow-up: the non-constant part of the state possesses a bounded absorbing set whose radius is independent of  $n$  and of the initial condition. Only the spatial mean can drift, and at most linearly—which is exactly the mode the mass penalty controls.

## 471 F Vacuity of the along-trajectory residual

472 Physics-informed training conventionally penalises the PDE residual evaluated on the model’s own  
 473 trajectory. The following makes precise why that penalty is uninformative for the present architec-  
 474 ture, and motivates the steady-state formulation adopted instead.

475 **Proposition 13** (The trajectory residual is an integrator diagnostic). *Define  $R_{\Delta t}(\mathbf{z}; \theta) :=$   
 476  $\Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_{\theta}(\mathbf{z}))$ . Then for every  $\theta$  in a bounded parameter set and every suffi-  
 477 ciently regular  $\mathbf{z}$ ,*

$$\|R_{\Delta t}(\mathbf{z}; \theta)\| = \Delta t \cdot \left\| \frac{1}{2}[\mathcal{A}_{\mathbf{D}}, \mathcal{F}_{\theta}]\mathbf{z} \right\| + O(\Delta t^2), \quad (9)$$

478 where  $\mathcal{F}_{\theta}\mathbf{z} := f_{\theta}(\mathbf{z})$  and  $[\cdot, \cdot]$  is the commutator. In particular  $R_{\Delta t} \rightarrow 0$  as  $\Delta t \rightarrow 0$  uniformly in  $\theta$ ,  
 479 and  $R_{\Delta t}$  does not depend on the observed data.

480 *Proof sketch.* Expanding  $S_{\Delta t}$  by BCH and subtracting the exact generator leaves the leading com-  
 481 mutator term at order  $\Delta t$ ; all data-dependent quantities cancel because  $S_{\Delta t}$  and the generator are  
 482 evaluated at the same state  $\mathbf{z}$ .  $\square$

483 Two consequences follow. First, minimising  $\|R_{\Delta t}\|^2$  cannot fit the model to observations, since the  
 484 observations do not appear in (9); it drives  $\theta$  towards operators whose diffusive and reactive parts  
 485 commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is  $O(\Delta t)$   
 486 and therefore vanishes in the very limit in which the discretisation is most faithful. The empirical  
 487 scaling is confirmed in Table 2 (observed order 1.00).

488 The constraint retained instead is the *steady-state* residual (3), which asks that the measured con-  
 489 figuration be a fixed point of the learned operator. This does involve the data, through  $\mathbf{z}_T$ , and it  
 490 constrains  $\mathbf{D}_{\theta}$  and  $f_{\theta}$  jointly: the set of pairs satisfying it is a proper subset of parameter space,  
 491 characterised next.

## 492 G Non-identifiability from a single snapshot

493 The following formalises the central epistemic limitation of the setting: an operator fitted to one  
 494 stationary field is determined only on a measure-zero subset of its own domain.

495 **Theorem 14** (Latent-space non-identifiability). *Let  $\mathbf{z}^* \in C^2(\Omega; \mathbb{R}^C)$  with  $\Omega \subset \mathbb{R}^2$  open, and  
 496 suppose  $C \geq 3$ . Fix any  $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$  and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^*) := \left\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) : \mathcal{A}_{\mathbf{D}}\mathbf{z}^*(x) + f(\mathbf{z}^*(x)) = 0 \quad \forall x \in \Omega \right\}. \quad (10)$$

497 *If  $\mathcal{F}(\mathbf{D}, \mathbf{z}^*) \neq \emptyset$  then it is an infinite-dimensional affine subset of  $C^1(\mathbb{R}^C; \mathbb{R}^C)$ . Specifically, for  
 498 every  $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$  vanishing on  $\mathcal{R} := \mathbf{z}^*(\Omega)$  and every  $f \in \mathcal{F}$ , one has  $f + \varphi \in \mathcal{F}$ , and the  
 499 space of such  $\varphi$  is infinite-dimensional.*

500 *Proof.* The constraint restricts  $f$  only through its values on  $\mathcal{R} = \mathbf{z}^*(\Omega)$ ; hence  $f + \varphi$  satisfies it  
 501 whenever  $\varphi|_{\mathcal{R}} \equiv 0$ . It remains to show that the space of such  $\varphi$  is infinite-dimensional. Since  $\mathbf{z}^*$   
 502 is  $C^1$  on a 2-dimensional domain,  $\mathcal{R}$  is a countable union of Lipschitz images of compact subsets  
 503 of  $\mathbb{R}^2$  and therefore has Hausdorff dimension at most 2. As  $C \geq 3$ ,  $\dim_H(\mathcal{R}) \leq 2 < C$ , so  $\mathcal{R}$   
 504 is Lebesgue-null in  $\mathbb{R}^C$  and has empty interior. Its complement therefore contains an open ball  $B$ ;  
 505 choosing countably many disjoint balls  $B_i \subset B$  and bump functions  $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$  yields a  
 506 linearly independent family, all vanishing on  $\mathcal{R}$ .  $\square$

507 **Corollary 15** (What a stationarity fit can determine). *Under the hypotheses of Theorem 14, the  
 508 steady-state loss (3) constrains  $f_{\theta}$  only on  $\mathcal{R}$ , a set of Lebesgue measure zero in  $\mathbb{R}^C$ . Any statement  
 509 about  $f_{\theta}$  off  $\mathcal{R}$ —in particular any counterfactual obtained by displacing the latent state away from  
 510 the observed manifold—is determined by the inductive bias of the architecture and the regularisers,  
 511 not by the data.*



Corollary 15 has a direct experimental consequence. The in-silico perturbation of Section 5.5 displaces  $\mathbf{z}$  along a latent direction and integrates forward; by construction this evaluates  $f_\theta$  at states off  $\mathcal{R}$ , where Theorem 14 shows the data exert no constraint. The null outcome reported there is therefore consistent with the theory rather than merely an empirical disappointment: a single stationary snapshot does not contain the information such a counterfactual would require. Breaking the degeneracy requires observations that visit a  $C$ -dimensional neighbourhood of  $\mathcal{R}$ —dense temporal sampling, or interventional data.

## H Linear stability and Turing analysis

Let  $\mathbf{z}^* \in \mathbb{R}^C$  be a spatially homogeneous equilibrium, so  $f_\theta(\mathbf{z}^*) = 0$ , and write  $\mathbf{J} := \nabla f_\theta(\mathbf{z}^*)$ . Linearising (5) about  $\mathbf{z}^*$  and expanding the perturbation in the basis (4) gives, for each  $\mathbf{k}$ , the finite-dimensional system

$$\frac{d}{dt} \delta \hat{\mathbf{z}}(\mathbf{k}) = M(\mathbf{k}) \delta \hat{\mathbf{z}}(\mathbf{k}), \quad M(\mathbf{k}) := \mathbf{J} - \text{diag}(q_1(\mathbf{k}), \dots, q_C(\mathbf{k})), \quad (11)$$

with  $q_c$  as in (6). Define the growth rate  $\mu(\mathbf{k}) := \max \text{Re spec } M(\mathbf{k})$  and its angular maximum  $\mu^*(k) := \max_{|\mathbf{k}|=k} \mu(\mathbf{k})$ ; the latter is required because anisotropy makes  $q_c$  direction-dependent.

**Definition 16** (Turing instability). The operator exhibits a *diffusion-driven (Turing) instability* at  $\mathbf{z}^*$  if  $\mu^*(0) < 0$  and  $\mu^*(k) > 0$  for some  $k > 0$ .

The first condition states that the reaction alone is stable; the second that diffusion destabilises a band of finite wavelengths. The selected wavelength is  $\lambda^* = 2\pi/k^*$  with  $k^* = \arg \max_k \mu^*(k)$ , converted to microns by the stored coordinate scale.

**Proposition 17** (Pure diffusion cannot pattern). If  $f_\theta \equiv 0$  then  $\mu^*(k) = -\min_c \min_{|\mathbf{k}|=k} q_c(\mathbf{k}) \leq -\lambda_{\min}(\mathbf{D})k^2 \leq 0$ , strictly decreasing in  $k$ . No Turing instability exists.

*Proof.*  $M(\mathbf{k}) = -\text{diag}(q_c(\mathbf{k}))$  is diagonal with real non-positive entries; its largest eigenvalue is  $-\min_c q_c(\mathbf{k})$ , and  $q_c(\mathbf{k}) \geq \lambda_{\min} |\mathbf{k}|^2$  by Lemma 4.  $\square$

**Proposition 18** (Two-channel criterion). Let  $C = 2$  with isotropic tensors  $\mathbf{D}_c = d_c I$ ,  $\mathbf{J} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , and write  $\sigma = |\mathbf{k}|^2$ . Then  $\mu(\mathbf{k}) > 0$  for some  $\sigma > 0$  while  $\mu(0) < 0$  if and only if

$$a + d < 0, \quad ad - bc > 0, \quad d_2 a + d_1 d > 2\sqrt{d_1 d_2 (ad - bc)}. \quad (12)$$

*Proof sketch.*  $M$  has trace  $a + d - (d_1 + d_2)\sigma$  and determinant  $h(\sigma) := d_1 d_2 \sigma^2 - (d_2 a + d_1 d)\sigma + (ad - bc)$ . The first two conditions give stability at  $\sigma = 0$ ; the trace is then negative for all  $\sigma > 0$ , so instability requires  $h(\sigma) < 0$  for some  $\sigma > 0$ . The minimum of the convex parabola  $h$  is attained at  $\sigma^* = (d_2 a + d_1 d)/(2d_1 d_2)$  and is negative precisely under the third condition.  $\square$

Condition (12) requires the two channels to have *unequal* diffusivities of opposite-signed feedback—the classical short-range-activation / long-range-inhibition requirement. The operators fitted in Section 5 satisfy the first two conditions but not the third: the dispersion curve is monotonically decreasing (Fig. 5c), and  $\mu^*(k) < 0$  for all  $k > 0$  in every run.

**Remark 19** (Why the negative result is expected). The training objective (3) rewards making the *observed* configuration stationary. A Turing-unstable operator does the opposite: it renders the homogeneous state unstable so that structure grows spontaneously. Since the observed field is treated as a fixed point rather than as the outcome of an instability, the fitted operator is driven towards dissipativity. Detecting pattern-forming instability would require observations away from steady state, consistent with Corollary 15.

## I Discretisation invariance of the encoder

The encoder must map measurements on an arbitrary point set to a field. Let  $\mu$  be the sampling measure on  $\Omega$  and consider the kernel-integral operator

$$(\mathcal{K}v)(x) = \int_{\Omega} \kappa_{\theta}(y - x) \odot v(y) d\mu(y), \quad (13)$$

of which the graph layer is the empirical counterpart  $(\mathcal{K}_N v)(x_i) = |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_{\theta}(x_j - x_i) \odot v(x_j)$ .

**Proposition 20** (Monte-Carlo consistency). *If  $\{x_j\}_{j=1}^N$  are i.i.d. draws from  $\mu$  and  $\kappa_{\theta} \odot v \in L^2(\mu)$ , then for each  $x$ ,  $\mathbb{E}[(\mathcal{K}_N v)(x)] = (\mathcal{K}v)(x)$  and  $\text{Var}[(\mathcal{K}_N v)(x)] = O(N^{-1})$ ; consequently  $\|\mathcal{K}_N v - \mathcal{K}v\|_{L^2(\mu)} = O_P(N^{-1/2})$ .*

*Proof.* The empirical operator is a sample mean of i.i.d. terms with finite second moment; unbiasedness and the  $O(N^{-1})$  variance are immediate, and the norm bound follows by Chebyshev and Fubini.  $\square$

Proposition 20 is the sense in which the encoder is discretisation-invariant: the population operator (13) depends on the geometry through  $\kappa_{\theta}(y - x)$  alone and not on the sampling pattern, and the empirical operator converges to it as the sampling density increases, whatever the lattice. **This is an asymptotic statement about the encoder, and it is worth emphasising that it does not imply transfer of the fitted operator.** The transfer experiments of Section 5 show empirically that it does not:  $\mathbf{D}_{\theta}$  and  $f_{\theta}$  are fitted to one tissue’s stationary field, and Theorem 14 indicates they are unconstrained away from that field’s range. Discretisation invariance of the *architecture* and portability of the *fit* are distinct properties, and only the former is established here.

**Proposition 21** (Splatting consistency). *Let  $w_{\sigma}(r) = e^{-r^2/2\sigma^2}$  and define the Nadaraya–Watson field  $F_{\sigma}(u) = \sum_i w_{\sigma}(\|u - x_i\|)v_i / \sum_i w_{\sigma}(\|u - x_i\|)$ . If  $v_i = v(x_i)$  for  $v \in C^2$  and the sampling density  $p$  is bounded away from zero, then  $F_{\sigma}(u) = v(u) + O(\sigma^2) + O_P((N\sigma^2)^{-1/2})$ .*

*Proof sketch.* Standard kernel-regression bias–variance decomposition in two dimensions: the bias term is  $\frac{\sigma^2}{2}(\Delta v + 2\nabla v \cdot \nabla \log p) + o(\sigma^2)$  and the variance is  $O((N\sigma^2)^{-1})$ .  $\square$

The learnable bandwidth  $\sigma$  therefore trades a smoothing bias against sampling noise. Since the bias term is proportional to  $\Delta v$ , splatting contributes a further diffusive smoothing on top of the operator—a mechanism that plausibly compounds the over-smoothing reported in Section 5, and one that the present work does not disentangle.

## J Complexity

Let  $N$  be the number of measured locations,  $G$  genes,  $C$  latent channels,  $H \times W$  the lattice,  $E$  the number of graph edges and  $T$  the integration horizon. The costs per forward pass are: gene projection  $O(NGd)$ ; graph layers  $O(LEd)$ ; splatting  $O(Nr^2C)$  for radius  $r$ ; spectral blocks  $O(BCHW \log HW)$ ; operator  $O(TCHW \log HW)$ ; decoding  $O(NCd + NGd)$ . The operator term is therefore linear in the horizon and quasi-linear in the lattice, and for the configurations used here ( $C=32$ ,  $H=W=64$ ,  $T=8$ ) it is under 4% of the forward cost—consistent with the parameter count of Section 6, where the operator holds 0.43% of the weights yet accounts for the entire margin over the strongest baseline.

## K Reproducibility

All datasets download automatically with a recorded SHA256 manifest; every table and figure in this paper is regenerated from run artifacts by a single command; configurations, seeds and checkpoints are released.

Table 3: Ablations on the mouse-brain Visium section. Each variant removes exactly one ingredient with all else held fixed.  $\Delta r$  is the change in held-out Pearson  $r$  relative to the full model. The *– dynamics* row is the critical control: it disables the operator entirely ( $T=0$ ) while leaving its parameters allocated, so the two rows are exactly parameter-matched. **This table uses a smaller training budget than Table 1** (fewer epochs and seeds); it is internally consistent, but absolute values are not comparable across the two tables.

| variant                   | params | Pearson $r \uparrow$                | $\Delta r$ |
|---------------------------|--------|-------------------------------------|------------|
| Full model                | 2.20M  | <b><math>0.233 \pm 0.005</math></b> | –          |
| – dynamics ( $T=0$ )      | 2.20M  | $0.227 \pm 0.001$                   | -0.007     |
| – diffusion term          | 2.20M  | $0.226 \pm 0.004$                   | -0.007     |
| – reaction term           | 2.19M  | $0.227 \pm 0.001$                   | -0.006     |
| – PDE constraints         | 2.20M  | $0.233 \pm 0.005$                   | -0.000     |
| – biological regularisers | 2.20M  | $0.233 \pm 0.005$                   | -0.000     |
| isotropic $D$             | 2.20M  | $0.229 \pm 0.004$                   | -0.004     |
| state-dependent $D$       | 2.20M  | $0.236 \pm 0.004$                   | +0.003     |
| discrete GNN operator     | 0.61M  | $0.223 \pm 0.001$                   | -0.011     |
| latent $C=8$              | 1.07M  | $0.213 \pm 0.005$                   | -0.021     |
| latent $C=16$             | 1.30M  | $0.220 \pm 0.002$                   | -0.013     |
| latent $C=64$             | 5.77M  | $0.237 \pm 0.000$                   | +0.004     |

Table 4: Processed spatial transcriptomics datasets. Locations and genes are counts after QC and gene selection; extent is the physical bounding box of the section.

| dataset             | technology                        | organism | locations | genes | extent ( $\mu\text{m}$ ) | resolution                 |
|---------------------|-----------------------------------|----------|-----------|-------|--------------------------|----------------------------|
| merfish_allen_31    | MERFISH, 500-gene pane            | Mus      | 91,951    | 502   | $9837 \times 6094$       | single cell                |
| merfish_allen_40    | MERFISH, 500-gene pane            | Mus      | 104,461   | 502   | $10000 \times 6961$      | single cell                |
| merfish_allen_50    | MERFISH, 500-gene pane            | Mus      | 92,787    | 501   | $8806 \times 6604$       | single cell                |
| mosta_embryo_E10.5  | Stereo-seq, DNA nanoba            | Mus      | 18,360    | 2,099 | $3500 \times 4950$       | bin50 ( 25 $\mu\text{m}$ ) |
| mosta_embryo_E9.5   | Stereo-seq, DNA nanoba            | Mus      | 5,898     | 2,069 | $1975 \times 2650$       | bin50 ( 25 $\mu\text{m}$ ) |
| perturb_norman      | Perturb-seq (CRISPRa),            | Homo     | 111,659   | 2,073 | –                        | single cell                |
| visium_human_breast | Visium (55 $\mu\text{m}$ spots, h | Homo     | 4,869     | 2,087 | $6395 \times 6750$       | spot (multi-cell)          |
| visium_mouse_brain  | Visium (55 $\mu\text{m}$ spots, l | Mus      | 2,691     | 2,079 | $4822 \times 6379$       | spot (multi-cell)          |
| xenium_mouse_brain  | Xenium In Situ, 248-ge            | Mus      | 36,362    | 248   | $5212 \times 3128$       | single cell                |

Table 5: In-silico morphogen source (*bead implant*). The latent perturbation direction is defined from a random half of each pathway’s genes; the table reports where the *held-out* half ranks in the response magnitude distribution (lower is stronger; 50% is chance).  $p$  is a permutation test over random gene sets of matched size. Response half-width is the distance at which the predicted response falls to half its peak.  $*p < 0.05$ .

| pathway | genes | held-out rank (%) | null (%) | $p$   | half-width ( $\mu\text{m}$ ) |
|---------|-------|-------------------|----------|-------|------------------------------|
| BMP     | 22    | $48.6 \pm 6.0$    | 49.7     | 0.475 | $575 \pm 154$                |
| FGF     | 19    | $72.6 \pm 0.8$    | 50.3     | 0.992 | $618 \pm 58$                 |
| SHH     | 15    | $56.6 \pm 6.7$    | 50.4     | 0.698 | 466                          |
| WNT     | 26    | $56.4 \pm 3.9$    | 49.9     | 0.762 | $649 \pm 143$                |

Table 6: Cross-tissue, cross-species transfer: adult mouse brain  $\rightarrow$  human breast carcinoma. Shared gene vocabulary: 390 genes. Mean  $\pm$  s.d. over seeds.

| model                                  | Pearson $r \uparrow$ | RMSE $\downarrow$ | SSIM $\uparrow$   |
|----------------------------------------|----------------------|-------------------|-------------------|
| <i>source (in-domain)</i>              |                      |                   |                   |
| STAGATE-style                          | $0.212 \pm 0.001$    | $0.249 \pm 0.000$ | $0.358 \pm 0.001$ |
| NMO (ours)                             | $0.209 \pm 0.004$    | $0.251 \pm 0.003$ | $0.365 \pm 0.004$ |
| <i>target (zero-shot)</i>              |                      |                   |                   |
| STAGATE-style                          | $0.093 \pm 0.001$    | $0.397 \pm 0.002$ | $0.266 \pm 0.002$ |
| NMO (ours)                             | $0.033 \pm 0.012$    | $0.409 \pm 0.005$ | $0.219 \pm 0.017$ |
| <i>target (decoder-only fine-tune)</i> |                      |                   |                   |
| STAGATE-style                          | $0.117 \pm 0.007$    | $0.393 \pm 0.000$ | $0.365 \pm 0.003$ |
| NMO (ours)                             | $0.041 \pm 0.019$    | $0.406 \pm 0.004$ | $0.282 \pm 0.017$ |
| <i>target (in-domain oracle)</i>       |                      |                   |                   |
| STAGATE-style                          | $0.147 \pm 0.001$    | $0.375 \pm 0.000$ | $0.427 \pm 0.001$ |
| NMO (ours)                             | $0.136 \pm 0.001$    | $0.397 \pm 0.002$ | $0.329 \pm 0.003$ |
| <i>target (training-mean floor)</i>    |                      |                   |                   |
| mean                                   | $-0.000 \pm 0.000$   | $0.404 \pm 0.002$ | $0.254 \pm 0.001$ |

Table 7: Cross-resolution transfer: 55  $\mu\text{m}$  Visium spots  $\rightarrow$  single-cell Xenium. Shared gene vocabulary: 248 genes. Mean  $\pm$  s.d. over seeds.

| model                               | Pearson $r \uparrow$ | RMSE $\downarrow$ | SSIM $\uparrow$   |
|-------------------------------------|----------------------|-------------------|-------------------|
| <i>source (in-domain)</i>           |                      |                   |                   |
| NMO (ours)                          | $0.256 \pm 0.009$    | $0.246 \pm 0.002$ | $0.384 \pm 0.007$ |
| STAGATE-style                       | $0.242 \pm 0.001$    | $0.248 \pm 0.001$ | $0.364 \pm 0.001$ |
| <i>target (zero-shot)</i>           |                      |                   |                   |
| STAGATE-style                       | $0.282 \pm 0.000$    | $1.765 \pm 0.001$ | $0.208 \pm 0.001$ |
| NMO (ours)                          | $0.139 \pm 0.015$    | $1.863 \pm 0.026$ | $0.125 \pm 0.004$ |
| <i>target (in-domain oracle)</i>    |                      |                   |                   |
| NMO (ours)                          | $0.259 \pm 0.006$    | $1.752 \pm 0.007$ | $0.556 \pm 0.006$ |
| STAGATE-style                       | $0.256 \pm 0.001$    | $1.760 \pm 0.002$ | $0.569 \pm 0.001$ |
| <i>target (training-mean floor)</i> |                      |                   |                   |
| mean                                | $0.000 \pm 0.000$    | $1.872 \pm 0.000$ | $0.101 \pm 0.001$ |

591 **L Ablation table**

592 **M Dataset inventory**

593 **N In-silico morphogen source: per-pathway results**

594 **O Cross-tissue transfer**

595 **P Cross-resolution transfer**

596 **Q Perturb-seq consistency of the reaction Jacobian**

597 **R Developmental forecasting: full results**

598 Every experiment so far fits a *stationary* operator, because the data are snapshots. The Stereo-seq  
599 developmental series is the single dataset here with a real time axis, so it is the only setting in which  
600 the operator can act as a forward-time predictor. The E9.5 field is encoded, integrated for  $T$  operator  
601 steps, and the decoded prediction scored against the measured E10.5 field (Table 9).

Table 8: Agreement between the model’s counterfactual gene–gene responses and measured CRISPRa responses from Norman et al. (2019). **This is an out-of-context probe:** the Perturb-seq data are non-spatial K562 cells while the operator is fitted to human breast tissue, so it tests whether the reaction module encodes generic transcriptional coupling, not whether it recovered tissue-specific signalling. Null is a within-perturbation label permutation;  $p$  is a Wilcoxon signed-rank test against that null.

| model         | #perturb. | Spearman $\rho$    | null   | % positive | $p$   |
|---------------|-----------|--------------------|--------|------------|-------|
| STAGATE-style | 7         | $0.010 \pm 0.000$  | +0.004 | 64%        | 0.633 |
| NMO (ours)    | 7         | $-0.018 \pm 0.009$ | -0.009 | 57%        | 0.844 |

Table 9: Developmental forecasting on Stereo-seq: the E9.5 field is encoded, integrated forward for  $T$  operator steps, and scored against the measured E10.5 field. **Consecutive stages are different embryos**, so there is no cell-to-cell correspondence and the comparison is made at the level of the rasterised field after isotropic normalisation to a common frame; it inherits the error of that coarse registration. *Persistence* predicts  $E10.5 = E9.5$  and is the reference the operator must beat for its dynamics to carry temporal information.

| model                     | horizon | field $r \uparrow$ | field RMSE $\downarrow$ | genes with $r > 0$ |
|---------------------------|---------|--------------------|-------------------------|--------------------|
| Persistence (no dynamics) | –       | 0.064              | 0.346                   | 61%                |
| Spatial-mean predictor    | –       | 0.000              | 0.257                   | 4%                 |
| NMO                       | $T=0$   | $0.082 \pm 0.001$  | 0.320                   | 74%                |
| NMO                       | $T=2$   | $0.089 \pm 0.000$  | 0.305                   | 72%                |
| NMO                       | $T=4$   | $0.092 \pm 0.000$  | 0.297                   | 73%                |
| NMO                       | $T=8$   | $0.094 \pm 0.000$  | 0.289                   | 72%                |
| NMO                       | $T=16$  | $0.095 \pm 0.001$  | 0.283                   | 72%                |
| NMO                       | $T=32$  | $0.090 \pm 0.003$  | 0.285                   | 70%                |

602 The comparison must be read carefully. **Consecutive MOSTA stages are different embryos**, not  
603 the same embryo imaged twice, so no cell-to-cell correspondence exists and none can be manu-  
604 factured by registration. Evaluation is therefore performed at the level of the rasterised field after  
605 isotropic normalisation to a common frame, and the score inherits the error of that coarse alignment.  
606 The reference to beat is *persistence* — predicting  $E10.5 = E9.5$ . If the learned dynamics carried no  
607 temporal information, NMO could not exceed it.

608 Persistence attains field-level  $r = 0.064$ , which is itself low and quantifies how much the embryo  
609 changes (and how crude the registration is) between these stages. Sweeping the horizon rather than  
610 tuning it, NMO reaches 0.095 at  $T = 16$ . The full sweep is reported so that the horizon is a stated  
611 quantity rather than a hyperparameter selected against the test field. This experiment carries the least  
612 statistical power of any reported here and is offered as an existence check on forward-time behaviour  
613 rather than as a forecasting benchmark.

## 614 S Counterfactual perturbation: full protocol and results

615 The computational analogue of a morphogen bead implant is performed: inject the latent direction  
616 that maximally raises a pathway’s decoded activity into a small disc, integrate the operator, and read  
617 out the spatial response. The design is deliberately falsifiable and split-half—the direction is defined  
618 from a random half of each pathway’s genes and enrichment is scored on the *held-out* half—with a  
619 permutation null over matched random gene sets.

620 **This test fails, and that is the primary outcome of the experiment.** Across 4 pathways and 2  
621 seeds, 0 reach significance; the smallest per-pathway  $p$  is 0.47 and the mean held-out rank is 59%  
622 against a 50% chance level, so if anything held-out pathway members respond *less* than random  
623 genes (Appendix N).

624 The result rules out a reading the correlations in Fig. 7b would otherwise invite. Moving the latent  
625 direction that most raises one half of a pathway does not preferentially move the other half, so  
626 those correlations are consistent with the field encoding coarse anatomy that pathway scores also  
627 track, rather than a pathway-specific representation. No claim is therefore made that NMO recovers  
628 signalling programmes. The experiment does establish a scale: the response decays to half its peak  
629 within  $577\ \mu\text{m}$ , matching the diffusion lengths read off  $\mathbf{D}_\theta$ . It is the *specificity*, not the range, that  
630 the test rejects.

631 Separately, the question of whether the reaction module encodes genuine transcriptional cou-  
632 pling is addressed by comparing the model’s counterfactual gene–gene responses against measured  
633 CRISPRa responses [Norman et al., 2019]. **This is an out-of-context probe and is treated as such:**  
634 Norman et al. profiled non-spatial K562 cells, whereas the operator here is fitted to tissue. Fitting is  
635 performed on the human breast section so that species and a proliferative context are at least shared;  
636 a permutation null is reported, as is the same quantity for every baseline.

637 **This probe is underpowered and is treated as inconclusive.** Only 7 of Norman’s single-gene  
638 perturbations fall inside the spatial model’s gene vocabulary (339 genes are shared overall): HVG  
639 selection on a breast-cancer section and a CRISPRa screen in K562 agree on very few targets. With  
640  $n = 7$  the test cannot separate absence of effect from absence of power. The observed value ( $\rho =$   
641  $-0.018 \pm 0.009$  against a null of  $-0.009$ , Wilcoxon  $p = 0.84$ ) is consistent with zero and supports no  
642 conclusion in either direction. It is reported because the experiment was pre-specified, not because  
643 it settled anything; a properly powered version would need a screen whose targets were chosen to  
644 overlap the spatial panel.

## 645 T Ablation figure

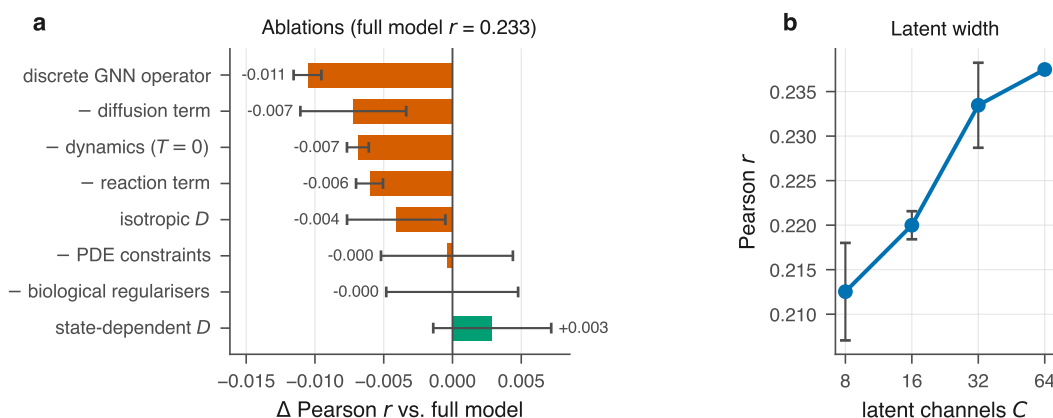


Figure 8: **Ablations.** (a) Change in held-out Pearson  $r$  when each ingredient is removed. (b) Latent-width sweep.

## 646 U Benchmark summary figure

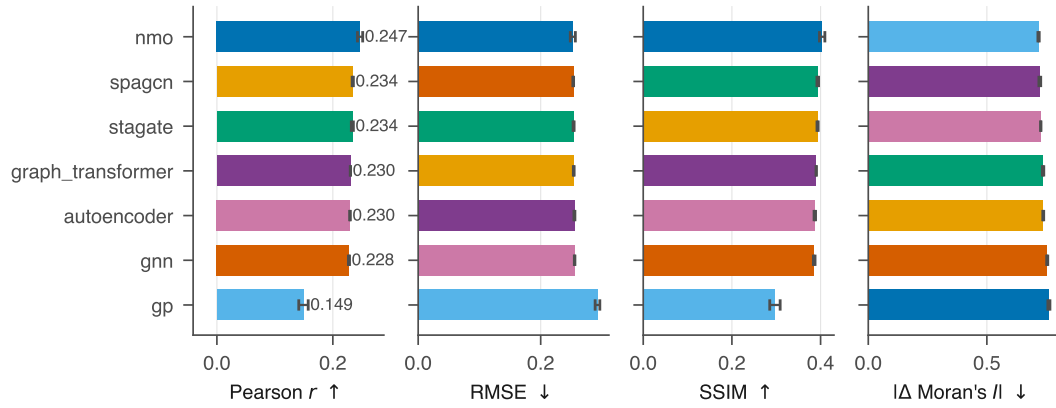


Figure 9: Benchmark of Table 1 across four complementary metrics. Moran’s  $I$  error (right) is the only axis on which NMO is not best.